AD 1 AERODROMES / HELIPORTS - INTRODUCTION
AD 1.1 AERODROME/HELIPORT - AVAILABILITY

1 General Conditions under which aerodromes / heliports and associated facilities are available for use

1.1 Introduction:

This section contains information an all aerodromes which are:-

a) ports of entry;
b) aerodromes for domestic use.

1.2 Aerodromes: Administration.

The administration of an aerodrome is the responsibility of the licensee.

1.3 Conditions of availability:

a) An aerodrome may be used by a particular aircraft, provided due account has been taken of the meteorological conditions obtaining at the time, provided the aircraft's performance and PCN is consistent with the physical characteristics of the aerodrome.
b) Pilots must ensure that prior permission is obtained from the licensee for the operation of all aircraft to and from aerodromes licensed in the private category.
c) The Defence Act 44 of 1957 states that no civilian aircraft may land at a military aerodrome without prior permission being obtained from :

The Chief of the Air Force

S.A Air Force Headquarters

Operations Room

TEL: (012) 312-2034

i) It must be pointed out, however, that even with prior permission, persons using such aerodromes do so at their own risk as the Defence Act also states that the Department of Defence shall not be liable to any person who makes use of the aerodrome or for loss or for damage resulting from damage to property caused by or arising out of the said defence service or property.

1.4 Applicable ICAO documents.

The standards and recommended practices contained in Annex 14 are applied with the exception of the differences noted hereunder.

1.5 Maintenance of aerodrome movement areas:

It is the responsibility of the relevant airport authority to maintain the aerodrome in a satisfactory condition.

1.6 Dissemination of information on condition of paved surfaces:

a) Information on surface condition of runways, taxiways and aprons will be published, when necessary, in NOTAM.

b) (b) At aerodromes where an ATS unit is established, if a runway is affected by standing water not associated with snow, slush or ice at any time during the approach of an aircraft for landing the depth and location of such standing water is notified by the aerodrome authority direct to the ATS at that aerodrome for transmission to the aircraft.

c) At certain Airports Company Airports tests have been carried out using a Mumeter to determine the surface friction characteristics, when wet, of paved runways, (ICAO Annex 14, Attachment B, para. 7.3.7 refers). The minimum Mumeter value for which a runway will be classified as “slippery when wet” is 0,5. If the friction of a runway is found to be below this value, then such information shall be promulgated by NOTAM.

1.7 Strength:

a) The assessment of the bearing strength of a runway is made complicated because the effective loads imposed by an aircraft of given weight vary with the number and layout of the wheels comprising the landing gear and the tyre pressure.

It is desirable that aircraft operators should be able to determine in advance for themselves, within reasonable limits, whether a particular runway is strong enough to accept the safe movement of a particular aircraft.

A system has been devised which expresses the bearing strength of a runway in a simple form and which provides sufficient information for the assessment in relation to this of the loads imposed by particular types of aircraft.

b) Pavement Classification Number (PCN):

The bearing strength of a runway is defined by a number (PCN).

If an operator proposes a movement or series of movements at an aerodrome, the aircraft PCN should be ascertained, as a function of single isolated wheel loading and tyre pressure.

If the resultant aircraft ACN is equal to or less than the published PCN for the runway it is intended to use, the bearing strength of that runway is sufficient for the proposed movements. Conversely, if the aircraft ACN is higher than the published PCN for the runway, the bearing strength may not be sufficient and the operator must obtain the prior permission of the aerodrome licensee for operations at that aerodrome.

The attention of all operators and pilots is drawn to the circumstances and conditions prevailing at many of the smaller aerodromes. Some of these aerodromes are unattended and, in spite of maintenance, holes and other damage may be caused by rodents overnight. Further, should an aerodrome become unserviceable for a short period after rain, unserviceability markers may not be displayed. Circumstances such as these will temporarily reduce the PCN of unpaved runways.

1.8 Aerodrome reference point (ARP)

The aerodrome reference point is the designated geographical location of an aerodrome. The point chosen is usually one situated near the geometric centre of the aerodrome.

1.9 Aerodrome reference temperature (ART).

The aerodrome reference temperature is the monthly mean of the mean daily temperature (24 hours) for the hottest month of the year (T1) (the hottest month being that which has the highest mean daily temperature) plus one-third of the difference between the monthly mean of the maximum daily temperature for the same month (T2) and the aforementioned monthly mean temperature (T1) i.e.Both T1, and T2 should be averaged over a period of years.

1.10 Rescue and fire fighting facilities.

Adequate rescue and fire fighting vehicles, equipment and personnel have been provided at most of the aerodromes available for use by international commercial air transport.

The scale of protection has been determined in accordance with the guidance given in Attachment C to Annex 14.

At the other major aerodromes the rescue and fire fighting equipment provided is usually suitable to their operating requirements.

1.11 Visual ground signals:

a) Of the visual ground signals contained in Appendix A to Annex 2, none have significance in the Republic of South Africa, with the exception of Para. 4.2.4 “Closed runways and taxiways”.

b) Closed manoeuvring areas, runways and taxiways.

1. Day markings.

Crosses of a single conspicuous colour, preferably white, displayed horizontally on the manoeuvring area shall indicate the limits of an area unfit for the movement of aircraft. In the case of runways and taxiways the crosses shall be displayed at each end of the unserviceable part and over the entire length of the unserviceable part at distances of not more than 300 M if the unserviceable part exceeds a distance of 300 M.

2. Night markings.

On manoeuvring areas used at night, unserviceable areas shall be delineated by fixed aviation red lights placed along the edge and at such corners of the area as may be necessary clearly to indicate the extent of the area, except that, where aircraft movements are limited to runways and taxiways, a runway or taxiway or portions thereof may be closed for taxiing purposes by placing aviation red lights across the entrance and exit of the section rendered unusable by reason of an unserviceable area, and by extinguishing the runway or taxiway lights between the delimiting red lights.

c) Traffic pattern indicator.

A traffic pattern indicator is used to indicate whether a right or left hand circuit should be followed in respect of a specific runway direction.

The indicator comprises, depending on the number of runways, a series of L markings around the wing cone circle.

The orientation of the markings coincide with that of the runways at the aerodrome where the indicator is displayed.

The L-shaped markings indicate whether a left or right hand circuit is in force. Thus when runway a or runway b is in use, a right hand circuit must be flown while, when runways c or d are in use, a left hand circuit is applicable.

1.12 **Stop bar Procedures**

Signs markings and lighting act as safety nets to prevent drivers or pilots from entering a runway incorrectly. Stop bars are provided at certain airports to assist in protecting the runway against a Runway Incursion. Airports using stop bars today typically equip the CAT II/III holding point as a compliment to the pavement marking and signage according to ICAO.

A stop bar consists of a row of unidirectional lights embedded in the pavement and spaced equally across the taxiway at the associated holding point, normally at right angles to the centerline. They show red towards an approaching aircraft when lit. Stop bars are sometimes installed in association with green lead-on lights which are a continuation of the taxiway centerline lighting beyond the stop bar. The lead-on lights are interlinked with the stop bar so that when the stop bar is red, the green centerline beyond the stop bar is unlit and vice versa.

Procedures for Stop bar Use

1) Pilots and Drivers shall never cross a red (lit) stop bar.
2) If instructed to cross a red stop bar pilots and drivers should challenge the ATC to switch off the stop bar or provide alternative routing.
3) Pilots and Drivers may only cross a deselected stop bar/holding point when the appropriate voice clearance has been issued by the ATC and read-back by the pilot/driver. Pilots and Drivers should be careful not to assume clearance to cross/enter a runway has been issued if the stop bar is not illuminated.
4) Should a stop bar fail ATC shall endeavour to provide an alternate clearance that does not route via the failed stop bar.
5) Where a stop bar has failed and there is no alternative routing available or the stop bar has failed at a runway entrance, a procedure shall be published either via NOTAM or promulgated in the AIP describing under what conditions pilots and drivers may be allowed to cross a lit stop bar. Should no such procedure

be published the pilots and drivers shall only cross the lit stop bar under the direction of a Follow-Me vehicle when ATC has given an explicit clearance for the Follow-Me vehicle to lead the aircraft onto/across the runway.

6) When a Follow-Me vehicle is used Pilots will be required to report the Follow-Me vehicle in sight to the ATC. Thereafter all Radio Transmissions will be directed to the Vehicle until the Driver of the Follow-Me vehicle reports standing down.
7) When issuing a Conditional Clearance, ATC should only extinguish the stop bar when the subject of the condition has passed and it is safe for the aircraft or vehicle to enter the runway at that entry point at that time.
8) Should a pilot or driver require clarification on a clearance issued they should stop before the stop bar and request clarification from the ATC prior to crossing any holding point. If in Doubt STOP and ask.